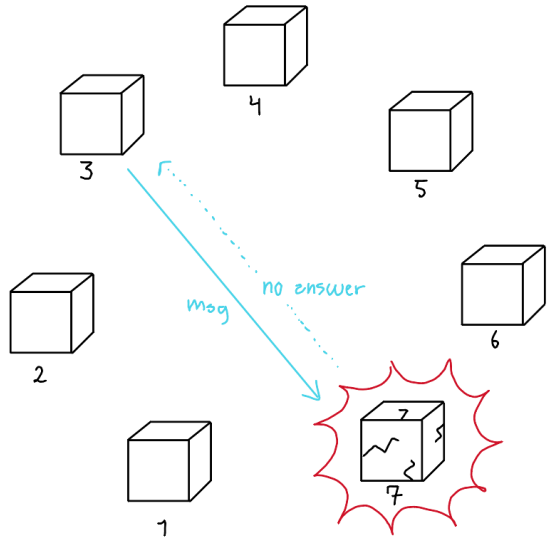
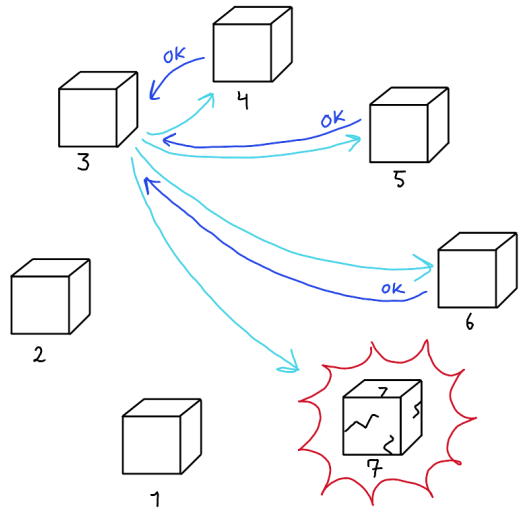
```
    net.deliver_all()
    net.run(4)
    assert all_agree_on(net, n - 1)
    by_type = net.messages_by_type()
    assert by_type["ELECTION"] == n * (n - 1) // 2
    assert by_type["OK"] == (n - 1) * (n - 2) // 2
    assert by_type["COORDINATOR"] == n - 1
    assert net.election_messages() == n * (n - 1)
```

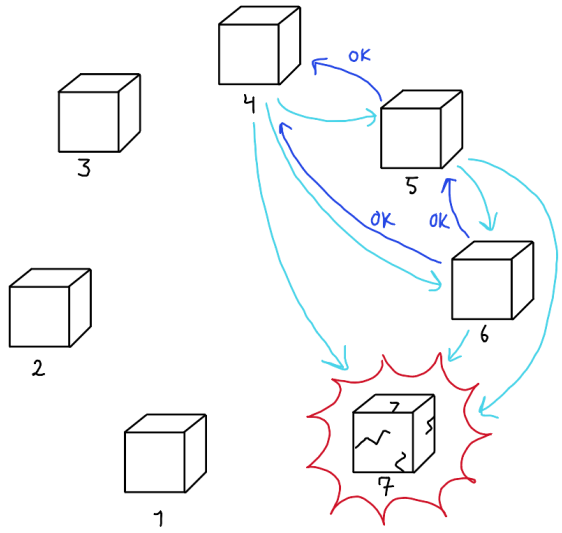
UML for Original Bully



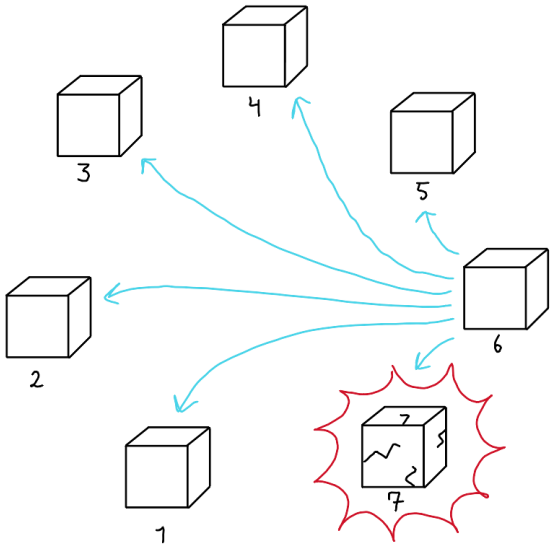
Leader with ID 7 is down.



Node with ID 3 sends election to all nodes with higher ID than itself.

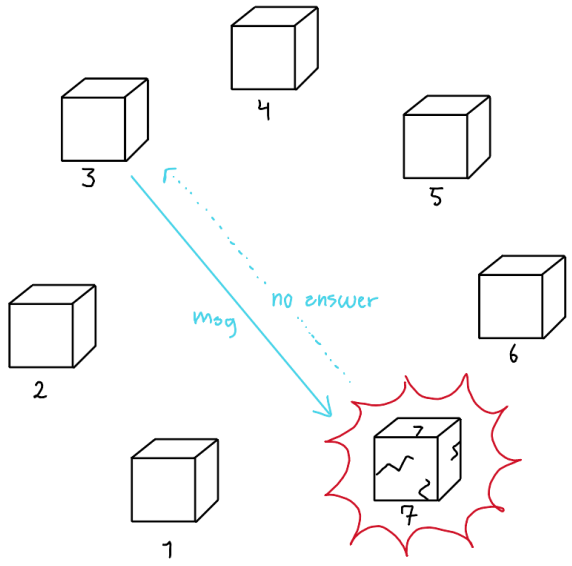


All nodes with higher ID's writes 'OK' and sends election to nodes with higher ID than themself.

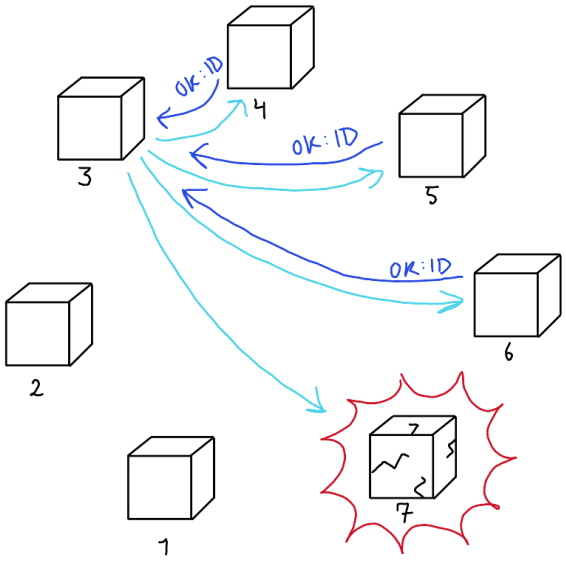


Highest node, that does not get 'OK' message becomes new leader. It sends coordinate message to all other nodes.

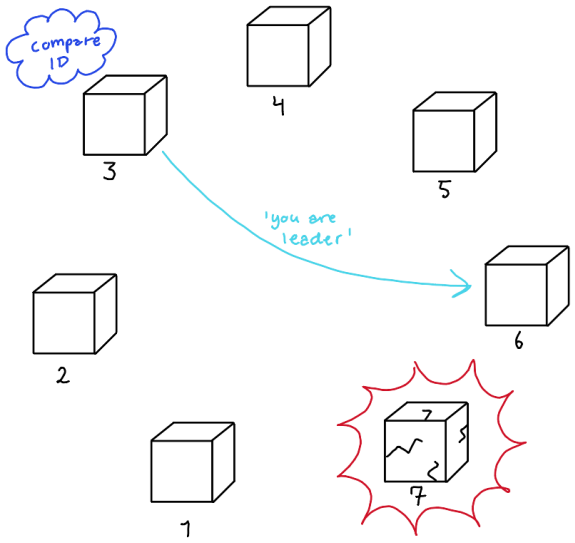
UML for Optimized Bully Election



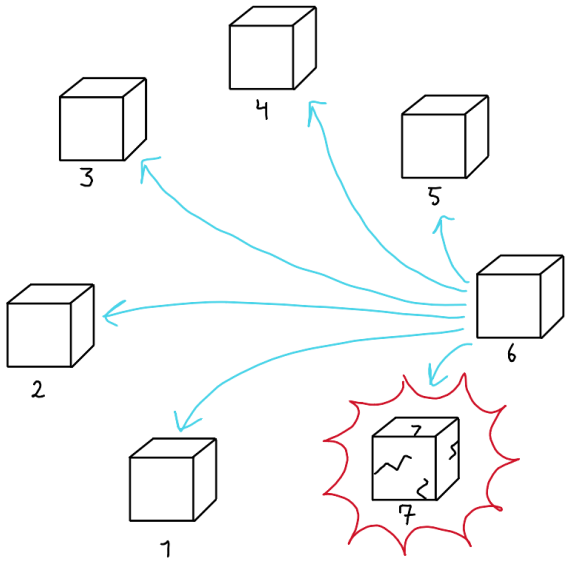
Leader with ID 7 is down.



Node with ID 3 sends election to all nodes with higher ID than itself.



Node with ID 3 compares ID from nodes that said ‘OK’ and sends leader message to node with highest ID



Highest node sends coordinate message to all other nodes.